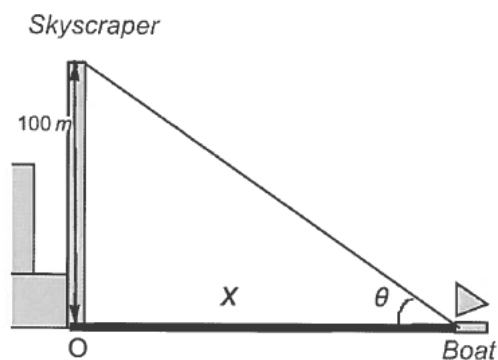


Problem 7

Problem 7

Problem 1 A boat sails directly toward a 100-meter skyscraper that stands on the edge of a harbor. The angular size θ of the building is the angle formed by lines from the top and bottom of the building to the observer on the boat (see figure below).



- Express the angle θ as the function of x , the distance of the boat from the building.
- Find the angular size, θ , when the boat is $x = 100\sqrt{3}\text{m}$ from the building.